the value stored in R1. In a pipelined structure, the second instruction will be reading the contents of R1 before the previous instruction has placed the value there. How could this potential problem be overcome?

18.03 The basic computer architectures

A well-established approach to describing different computer architectures is to consider the number of instruction streams and the number of data streams. This leads to four different types.

Single Instruction Stream Single Data Stream (SISD)

SISD is the typical arrangement found in early computers which was also adopted for the earliest microprocessors. The functioning is purely sequential with no parallelism. Figure 18.02 illustrates how individual elements in an array are manipulated using SISD.

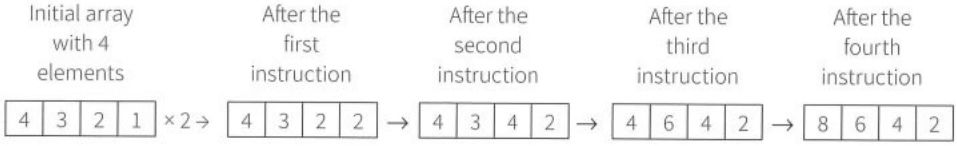


Figure 18.02 SISD sequential processing of array elements

The same instruction is used repeatedly to carry out the multiplication by two.

Single Instruction Stream Multiple Data Stream (SIMD)

SIMD can be considered as parallelism applied to the data stream. The difference when array elements are processed is shown in Figure 18.03.

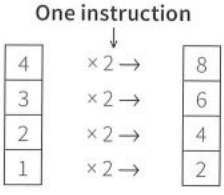


Figure 18.03 SIMD parallel processing of array elements

The architecture for SIMD is usually represented by a diagram such as Figure 18.04.

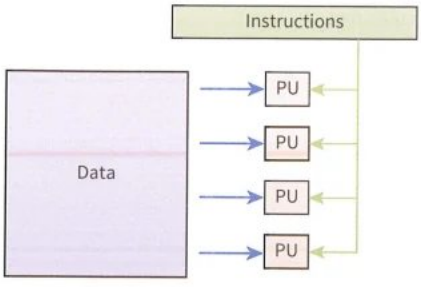


Figure 18.04 Schematic representation of the SIMD architecture

In Figure 18.04 the structure shows four data streams entering four individual components which are simultaneously supplied with the same instruction. The components are labelled PU (Processing Unit). Sometimes the components are labelled PE (Processing Element). Whatever name is used, the components are arithmetic logic units. One of the reasons for the different names is that there are options for how this architecture could be implemented.

TIP

Be careful to distinguish between the pipelining of instructions where different instructions are executed in parallel and SIMD when only one instruction is being simultaneously executed on different data streams.

One option has been used in computers that are called array or vector processors. These might have a parallel set of registers; one for each data stream. Alternatively, there would be a large register, perhaps with 64 or 128 bits which could store four data values at the same time. In this type of implementation, the parallelism is built into just one processor.

The alternative is the multi-core processor where four individual processors work in parallel. In this case each processor is likely to have its own dedicated cache memory to provide the data stream.

KEY TERMS

SISD: Single Instruction Stream Single Data stream; a single processor accessing one memory

SIMD: Single Instruction Stream Multiple Data stream; processing of parallel data input requiring one control unit instructing multiple processing units

MISD: Multiple Instruction Stream Single Data stream; does not exist in a single architecture

MIMD: Multiple Instruction Stream Multiple Data stream; multiple processors asynchronously processing parallel data input

Multiple Instruction Stream Single Data Stream (MISD)

MISD is not evidenced in any individual computer architecture design. One example where the approach could be implemented would be in a fault-tolerant system. The same data stream could be fed into two or more processors. The output would only be accepted if the same output was produced by all of the processors.

Multiple Instruction Stream Multiple Data Stream (MIMD)

The **MIMD** architecture is similar to that for the type of SIMD architecture shown in Figure 18.04 with more than one processing unit receiving the parallel data streams. The difference is that each Processing unit does not execute the same instruction. The multiple data stream can be provided by a suitably partitioned single memory. Each Processing Unit might have a dedicated cache memory.

Massively parallel computer systems

The **MIMD** architecture can be implemented in multicomputer systems known as massively parallel computers. These are the systems used by large organisations for computations involving highly complex mathematical processing. They are the latest type of 'supercomputer'. The major difference in architecture is that instead of having a bus structure

to support multiple processors there is a network infrastructure to support multiple computer units. The programs running on the different computers can communicate by passing messages using the network.

An alternative type of multicomputer system is cluster computing using PCs (sometimes referred to as a 'server farm').

Whatever the technology used, these systems have an extremely large number of individual processors working in parallel.

Extension Question 18.02

The IBM Sequoia and a Beowulf cluster are different examples of cluster computing. You might wish to carry out some research to get some details of their computing power.

18.04 Virtual machines

A virtual machine is software not hardware. The most usual type of virtual machine is the **system virtual machine** which is software that emulates the hardware of a real computer system.

KEY TERM

System virtual machine: the emulation of computer system hardware using software

Remember that when a virtual machine is not being used an application program requires support from an operating system in order for the program to run on the hardware. The principle of a virtual machine is that a process interacts directly with a software interface provided by an operating system. The logical structure for the operation of a system virtual machine is shown in Figure 18.05.

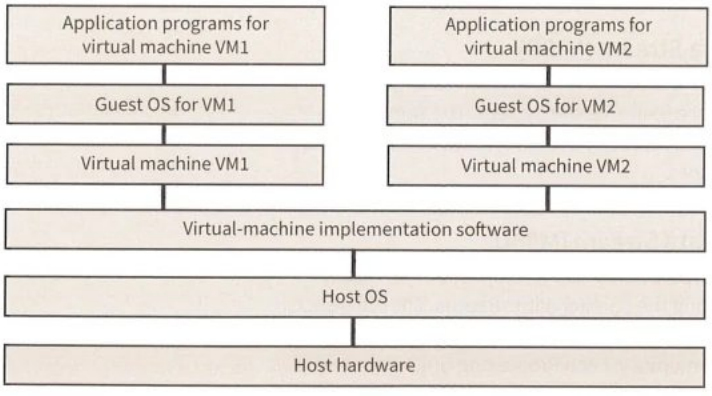


Figure 18.05 Logical structure for a system virtual machine implementation

The following points should be noted when looking at Figure 18.05.

1 The application programs are installed with the assistance of a guest OS. This guest OS will support the running application by interacting with the virtual machine as though it were the hardware that the guest OS would normally run on.